MATH 152 **EXAM IIA** NAME _____

Mrs. Bonny Tighe 50 points

7.7-11.7 TA_____ Wed 10/31/2018

PART II

1. (10 points) Complete the following

GEOMETRIC SERIES $\sum_{n=1}^{\infty} a_n (r)^{n-1}$ is _____ if $|r| \geq 1$

THE COMPARISON TEST Suppose $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are series with positive terms.

i) If $\sum_{n=1}^{\infty} b_n$ is convergent and a_n ____ b_n for all n, then $\sum_{n=1}^{\infty} a_n$ is also convergent

ii) If $\sum_{n=1}^{\infty} b_n$ is divergent and a_n ____ b_n for all n, then $\sum_{n=1}^{\infty} a_n$ is also divergent

THE LIMIT COMPARISON TEST

Suppose that $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are series with positive terms . If $\lim_{n\to\infty} \frac{a_n}{b_n} = c$ where c is _____, then either both series converge or both diverge.

TEST FOR DIVERGENCE **If** $\lim_{n\to\infty} a_n \neq$ _____ , then $\sum_{n=1}^{\infty} a_n$ is divergent

THE INTEGRAL TEST Suppose f is a continuous, positive, decreasing function on $[1, \infty)$ and let $a_n = f(n)$ then the series $\sum_{n=1}^{\infty} a_n$ is convergent if and only if _____is convergent.

P-SERIES $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is _____ if $p > 1$

RATIO TEST

If $\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = L$ $\begin{cases} L ____ 1 \text{ then } \sum a_n \text{ is absolutely convergent} \\ L ____ 1 \text{ then } \sum a_n \text{ is divergent} \\ L = 1 \text{ then the test is } _____ \end{cases}$

ROOT TEST

If $\lim_{n\to\infty} \sqrt[n]{|a_n|} = L$ and $\begin{cases} L ____ 1 \text{ then } \sum a_n \text{ is absolutely convergent} \\ L ____ 1 \text{ then } \sum a_n \text{ is divergent} \\ L = 1 \text{ then the root test is } _____ \end{cases}$

THE ALTERNATING SERIES TEST

If i) is $b_{n+1} \leq b_n$ for all n and ii) $\lim_{n\to\infty} b_n = 0$ Then $\sum_{n=1}^{\infty} (-1)^{n-1} b_n$ is _____

REMAINDER ESTIMATE FOR THE INTEGRAL TEST

Suppose $a_k = f(k)$ where f is a continuous, positive, decreasing function for $x \geq n$, and $\sum a_n$ is convergent . If $R_n = S - S_n$, then _____ $\leq R_n \leq$ _____

2. (10 points). Use the **Integral test** to determine if the series is divergent or convergent.

$$\sum_{n=0}^{\infty} \frac{e^n}{e^{2n}+1}$$

3. (10 points) Multiple Choice: Circle the correct letter

a) Find a sequence for $1, \frac{1 \cdot 5}{1 \cdot 2}, \frac{1 \cdot 5 \cdot 9}{1 \cdot 2 \cdot 3}, \frac{1 \cdot 5 \cdot 9 \cdot 13}{1 \cdot 2 \cdot 3 \cdot 4}, \ldots$ is

a) $a_n = \frac{4n-3}{n!}$

b) $a_n = \frac{(1 \cdot 5 \cdot 9 \cdots (4n-3))}{(n+1)!}$

c) $a_n = \frac{(1 \cdot 5 \cdot 9 \cdots (4n-3))}{n!}$

d) $a_n = \frac{4n-1}{(n+1)!}$

e) None of the above

b) $\sum_{n=1}^{\infty} \frac{(-5)^{n+1}}{3^{2n}}$ equals

a) $\frac{-45}{14}$

b) $\frac{-45}{4}$

c) $\frac{25}{14}$

d) $\frac{-25}{4}$

e) None of the above

c) Which of the following sequences is not convergent?

a) $a_n = \frac{4n-1}{2n+1}$

b) $a_n = \frac{5^n+2}{3^n}$

c) $a_n = \frac{3}{n!}$

d) $a_n = \frac{\arctan n}{\sqrt{n+2}}$

e) They are all convergent

d) Which test would you use to determine the convergence of the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^3+3n+1}$

a) Alternating Series Test
b) Integral Test
c) Limit Comparison Test
d) Ratio Test
e) None of the above tests

e) Which test would you use to determine the convergence of the series $\sum_{n=0}^{\infty} \frac{(-1)^n}{n!}$

a) Limit Comparison Test
b) Root Test
c) Integral Test
d) Ratio Test
e) None of the above

4. (5 points) True False – Write the word

a) The first test you try when determining the convergence of a series is the ratio test. _____

b) Given $a_n = n^2 + 2n + 1$, then $a_{n+1} = n^2 + 4n + 4$. _____

c) To determine the convergence of $\sum_{n=0}^{\infty} \frac{n-1}{n^2+n^3}$ I could use the Integral Test.

_____

d) If $\lim_{n\to\infty} a_n = 0$, then the series $\sum_{n=0}^{\infty} a_n$ is convergent. _____

e) If $\lim_{n\to\infty} a_n = 0$, then the sequence $\{a_n\}$ is convergent. _____

5. (10 points) Determine whether the series is convergent or divergent and if it is convergent **find the sum**.

a) $\sum_{k=1}^{\infty} \frac{1}{k^2+3k}$

b) $\sum_{k=1}^{\infty} \frac{3^k}{5^{k+1}}$

6. (10 points) Test the following series for absolute convergence, conditional convergence or divergence. State the test you are using and show all your work.

a) $\sum_{k=1}^{\infty} \frac{k+1}{\sqrt{k^5+3k^2+1}}$

b) $\sum_{n=1}^{\infty} \frac{(-1)^n}{1+2n}$

c) $\sum_{n=1}^{\infty} \frac{1 \cdot 4 \cdot 7 \cdot 10 \cdots (3n-2)}{5^n n!}$

MATH 152
Mrs. Bonny Tighe

EXAM IIA

50 points
7.7-11.7
PART I

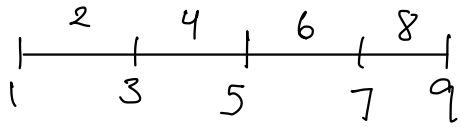
NAME _____

TA_____ Wed 10/31/2018

1. (10 points) Use (a) the Midpoint Rule (b) the Trapezoidal Rule and (c) Simpson's Rule to approximate the given integral with the specified value of n. Find the error bounds for the Midpoint rule. Suppose $|f''(x)| \le K$ *for* $a \le x \le b$, then

$\frac{9-1}{4} = 2$

$$|E_M| \le \frac{K(b-a)^3}{24n^2} \qquad \int_1^9 \frac{1}{x^2}\,dx, \quad n = 4$$



$$M_4 = 2\left[\frac{1}{4} + \frac{1}{16} + \frac{1}{36} + \frac{1}{64}\right]$$

$$T_4 = 1\left[1 + 2\left(\frac{1}{3^2}\right) + 2\left(\frac{1}{5^2}\right) + 2\left(\frac{1}{7^2}\right) + \frac{1}{9^2}\right]$$

$$S_4 = \frac{2}{3}\left[\frac{1}{1^2} + 4\left(\frac{1}{3^2}\right) + 2\left(\frac{1}{5^2}\right) + 4\left(\frac{1}{7^2}\right) + \frac{1}{9^2}\right]$$

$f(x) = x^{-2}$

$f'(x) = -2x^{-3}$

$f''(x) = 6x^{-4}$

$|6x^{-4}|$ largest when $x = 1$

$K = 6$

$$\frac{6(9-1)^3}{24(4)^2} = \frac{\cancel{6}(8)^3}{\cancel{24}_4(4)^2} = \frac{8^3}{4^3} = 2^3 = \boxed{8}$$

How large should we take *n* in order to guarantee that the Midpoint approximation for $\int_1^9 \frac{1}{x^2}dx$ is accurate to within 0.001?

$$\frac{8^3}{4n^2} \le \frac{1}{10^3} \qquad \frac{80^3}{4} \le n^2 \qquad \sqrt{\frac{80^3}{4}} \le n$$

next largest integer

2. (10 points) The given curve is rotated about the x-axis. Find the area of the resulting surface. $S = 2\pi \int_a^b radius\,arclength$ $\quad y = \sqrt{6+x}, \quad 3 \le x \le 4$

$f = y$

$\frac{dy}{dx} = \frac{1}{2\sqrt{6+x}}$

$\left(\frac{dy}{dx}\right)^2 = \frac{1}{4(6+x)}$

$$S = 2\pi \int_3^4 \sqrt{6+x}\,\sqrt{1 + \frac{1}{4(6+x)}}\,dx$$

$$S = 2\pi \int_3^4 \sqrt{6+x}\,\sqrt{\frac{4(6+x)+1}{4(6+x)}}\,dx$$

$$S = \cancel{2}\pi \int_3^4 \cancel{\sqrt{6+x}}\,\frac{\sqrt{25+4x}}{\cancel{2}\cancel{\sqrt{6+x}}} = \pi \int_3^4 \sqrt{25+4x} = \frac{\cancel{2}}{3}\cdot\frac{1}{\cancel{4}_2}\cdot\pi\left(25+4x\right)^{\frac{3}{2}}\Big]_3^4$$

$$\frac{\pi}{6}\left[(25+4(4))^{\frac{3}{2}} - (25+4(3))^{\frac{3}{2}}\right] = \frac{\pi}{6}\left[41^{\frac{3}{2}} - 37^{\frac{3}{2}}\right]$$

$$\frac{\pi}{6}\Big[$$

3. (10 points) Determine if the following improper integrals are convergent or divergent. If convergent, evaluate.

a) $\int_1^{\infty} x\, e^{-x}\, dx$

$$\lim_{t\to\infty} \int_1^t xe^{-x}dx \qquad u = x \quad dv = e^{-x}$$
$$du = dx \quad v = -e^{-x}dx$$

$$\lim_{t\to\infty} -xe^{-x} + \int_1^t e^{-x}dx$$

$$\lim_{t\to\infty} \left[-\frac{x}{e^x} - \frac{1}{e^x}\right]_1^t$$

$$\lim_{t\to\infty} \left[-\frac{t}{e^t} - \frac{1}{e^t} - \left(-\frac{1}{e} - \frac{1}{e}\right)\right]$$

$$= 0 - 0 + \frac{1}{e} + \frac{1}{e} = \boxed{\frac{2}{e}}$$

converges

b) $\int_1^3 \frac{x}{\sqrt{x^2-1}}\, dx = \int_1^3 x(x^2-1)^{-\frac{1}{2}}\, dx$

$u = x^2 - 1$

$du = 2x\, dx$

$\frac{1}{2}du = x\, dx$

$$\lim_{t\to 1^+} \cancel{\frac{1}{2}\cdot\frac{2}{1}}\left[\sqrt{x^2-1}\right]_1^3$$

$$\sqrt{9-1} - \sqrt{0}$$

$$\boxed{= \sqrt{8}}$$

convergent

4. (10 points) Find the length of the arc of the parabola

$$x = \frac{y^2}{2} - \frac{1}{4}\ln y \quad on \quad 1 \le y \le 2$$

5. (10 points) Multiple Choice – (2 points each) Circle the correct lettera) When using the comparison test for the improper integral $\int_2^{\infty} \frac{x}{\sqrt{x^3 - 2x + 3}}\,dx$ the inequality will be

a) $\cancel{\frac{1}{\sqrt{x^3}}} \le \frac{x}{\sqrt{x^3 - 2x + 3}}$

b) $\cancel{\frac{1}{\sqrt{x^3}}} \ge \frac{x}{\sqrt{x^3 - 2x + 3}}$

c) $\frac{1}{\sqrt{x}} \le \frac{x}{\sqrt{x^3 - 2x + 3}}$

d) $\frac{1}{\sqrt{x}} \ge \frac{x}{\sqrt{x^3 - 2x + 3}}$

e) None of the above

$x^3 - 2x + 3 \le x^3$

$x^{\frac{2}{2}} - 2x + 3 \le 0$

$\frac{x^{\frac{2}{2}}}{x^{\frac{3}{2}}} = \frac{1}{x^{\frac{1}{2}}} = \boxed{\frac{1}{\sqrt{x}}}$

$\frac{1}{2} \le 1$

divergent

$\ge$

b) The surface area for the function $y = \ln x$ when rotated about the y-axis on the interval $2 \le y \le 3$ is:

$r = x$

$x = e^y$

a) $2\pi \int_2^3 y\sqrt{1 + e^{2y}}\,dy$

b) $2\pi \int_2^3 e^y \sqrt{1 + e^{2y}}\,dy$

c) $2\pi \int_2^3 x\sqrt{1+\left(\frac{1}{x^2}\right)}\,dx$

d) $2\pi \int_2^3 \ln x\sqrt{1+\left(\frac{1}{x^2}\right)}\,dx$

e) None of the above

c) The K for the error bound for the midpoint formula for approximating $\int_1^4 \sqrt{x}\,dx$ with 4 subdivisions is:

$f(x)=\sqrt{x}$

$f'(x)=\frac{1}{2}x^{-\frac{1}{2}}$

$f''(x)=-\frac{1}{4}x^{-\frac{3}{2}}$ $\left|\frac{1}{4}\right|$

a) $K=-\frac{1}{4}$

b) $K=\frac{1}{2}$

c) $K=\frac{1}{4}$

d) $K=\frac{1}{32}$

e) None of the above

d) Which of the following is not an improper integral?

a) $\int_{-2}^{5}\frac{1}{\sqrt{x+2}}\,dx$

b) $\int_1^3\frac{x}{x^2+3x-4}\,dx$

c) $\int_1^3\frac{x}{x^2-3x+4}\,dx$

d) $\int_0^1\frac{1}{\ln x}\,dx$

e) They are all improper integrals

e) Represent $0.4\overline{7}$ as a series

$\frac{4}{10}+\sum_{n=0}^{\infty}\frac{7}{100}\left(\frac{1}{10}\right)^n$

a) $\frac{4}{10}+\sum_{n=0}^{\infty}\left(\frac{7}{10}\right)^n$

b) $\frac{4}{10}+\sum_{n=0}^{\infty}\frac{7}{100}\left(\frac{1}{10}\right)^n$

c) $\frac{4}{10}+\sum_{n=0}^{\infty}\frac{7}{10}\left(\frac{1}{100}\right)^n$

d) $\frac{4}{10} + \sum_{n=0}^{\infty} 7\left(\frac{1}{10}\right)^n$

e) None of the above

6. (5 points) True False – Write the word

a) The formula for Simpson's rule for approximation of a definite integral with four subdivisions is $\Delta x\left[f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + f(x_4)\right]$. False

b) This $\int_1^3 \frac{x}{\sqrt{x^3 + 2x - 3}}dx$ is an improper integral. True

c) The arc length of the function $y = \sqrt{x}$ on the interval $2 \le y \le 3$ is $\int_2^3 \sqrt{1 + \left(\frac{1}{4x}\right)}\,dx$

$x = y^2$ dy

False

d) The approximation for Trapezoid rule for a definite integral with four subdivisions is $\Delta x/2\left[f(x_1) + 2f(x_2) + 2f(x_3) + f(x_4)\right]$. False

e) The arc length for the function $y = e^{3x}$ on the interval $1 \le x \le 2$ is $\int_1^2 \sqrt{1 + \left(9e^{9x^2}\right)}dx$

dx

False

$\frac{dy}{dx} = 3e^{3x}$

$\left(\frac{dy}{dx}\right)^2 = 9e^{6x}$